

YEON-JI SONG

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EDUCATION

Seoul National University (SNU)

PhD in Interdisciplinary Program in Neuroscience & AI Institute

- Advisor: Prof. Byoung-Tak Zhang

2021.09 – 2027.09 (expected)

Seoul, South Korea

Hong Kong University of Science and Technology (HKUST)

BEng in Electronic and Computer Engineering

2017.09 – 2021.06

Clear Water Bay, Hong Kong

RESEARCH INTERESTS

Visual generation via physical concept grounding, including moving objects and camera dynamics, in a neurosymbolic way.

Dynamic scene understanding for novel view synthesis and reconstruction from blurry monocular inputs.

Object-centric learning in the physical world with time-static object appearance and time-varying object motion.

Robotics and embodied AI leveraging learned object and scene representations for robotics tasks in real-world settings.

PUBLICATIONS

*equal contribution, †corresponding author

Kinematics-Driven Gaussian Shape Deformation for Blurry Monocular Dynamic Scenes

Yeon-Ji Song, Kiyoungh Kwon, Junoh Lee, Jin-Hwa Kim[†], Byoung-Tak Zhang[†]

in *Proceedings of ICML 2026*

Relaxed Rigidity with Ray-based Grouping for Dynamic Gaussian Splatting

Junoh Lee, Junmyeong Lee, Yeon-Ji Song, Inhwon Bae, Jisu Shin, Hae-Gon Jeon[†], Jin-Hwa Kim[†]

Preprint

One-shot 3D Affordance Learning for Multi-stage Robotic Manipulation

Hyunseo Kim, Yeon-Ji Song, Minsu Lee[†], Byoung-Tak Zhang[†]

in *IEEE Access 2026*

OCK: Unsupervised Dynamic Video Prediction with Object-Centric Kinematics

Yeon-Ji Song, Jaein Kim^{*}, Suhyung Choi^{*}, Jin-Hwa Kim[†], Byoung-Tak Zhang[†]

in *Proceedings of ICCV 2025*

DBMovi-GS: Dynamic View Synthesis from Blurry Monocular Video via Sparse-Controlled Gaussian Splatting

Yeon-Ji Song, Jaein Kim, Byoungju Kim, Byoung-Tak Zhang[†]

in *Proceedings of CVPR 2025 Workshop on Neural Fields Beyond Conventional Cameras*

Continuous SO(3) Equivariant Convolution for 3D Point Cloud Analysis

Jaein Kim, Heebin Yoo, Dong-Sig Han, Yeon-Ji Song, Byoung-Tak Zhang[†]

in *Proceedings of ECCV 2024*

Unsupervised Visual Dynamics Learning with Multi-Object Kinematics

Yeon-Ji Song, Byoung-Tak Zhang[†]

in *Proceedings of KCC 2024*

Learning Object Appearance and Motion Dynamics with Object-Centric Representations

Yeon-Ji Song, Hyunseo Kim, Suhyung Choi, Jin-Hwa Kim[†], Byoung-Tak Zhang[†]

in *Proceedings of NeurIPS 2023 Workshop on Causal Representation Learning*

On Discovery of Local Independence over Continuous Variables via Neural Contextual Decomposition

Inwoo Hwang, Yunhyeok Kwak, Yeon-Ji Song, Byoung-Tak Zhang[†], Sanghack Lee[†]

in *Proceedings of CLeaR 2023*

PROFESSIONAL SERVICE

PROGRAM COMMITTEE MEMBER (REVIEWER)

- NeurIPS 2026, ICML 2026, ICCV 2025, WACV 2024

TECHNICAL MENTORING

- AI Youth Challenge (POSCO DX) 2023 – 2026
- Machine Learning and Computer Vision (Hyundai NGV) 2025
- Project XR: AI Chatbot (LognCoding) 2024

TEACHING ASSISTANT

Seminars in Neuroscience 1 & 2 (SNU)	2025.09 – 2026.06
Multimodal Generative AI Theories and Applications (SNU)	2025.09 – 2025.12
Multimodal Deep Learning Theories and Applications (SNU)	2024.09 – 2024.12
Artificial Intelligence (SNU)	2022.03 – 2022.06
New Computer Technology (SNU x HKUST)	2022.03 – 2022.06
ENGG2900D: ROV Community Project (HKUST)	2019.02 – 2022.05

SCHOLARSHIPS AND AWARDS

Samsung Industrial-Academic Scholarship	2025 – 2027
SNU Neuroscience Outstanding Poster Award	2025
KCC 2024 Best Presentation Paper Award	2024
RoboCup@Home DSPL 2nd Place	2022
HKUST Admission Scholarship	2017

WORK EXPERIENCE

Biointelligence Lab (SNU) <i>Undergraduate Research Intern</i> Designed and conducted research on Robotics and Reinforcement Learning.	2020.12 – 2021.04 <i>Seoul, South Korea</i>
Surromind (SNU) <i>Artificial Intelligence Research Engineer</i> Designed and implemented a Deep Learning model for Pose Estimation based on Detectron2.	2020.07 – 2020.09 <i>Seoul, South Korea</i>
Robocore AI <i>Robotics Software Engineer</i> - Created new solutions for temi robots, combining IOT products and the mobility of temi. - Performed GUI design, system design and solved real-life customer request with AI solution.	2020.06 – 2020.07 <i>Science Park, HongKong</i>
Codecrain Inc. <i>Full-stack Web Developer</i> - Developed a frontend web application along with a senior developer using React.js. - Implemented React.js and Node.js to enhance functionality and user experience.	2019.06 – 2019.09 <i>Seoul, South Korea</i>

PROJECTS

SNU-NAVER Hyperscale AI Center <i>Student Researcher</i> - Advisor: Jin-Hwa Kim (Leader of Generation Research at NAVER AI Lab) - Developed video generative models focused on 3D object and motion dynamics. - Published at ICCV 2025 and NeurIPS 2023 workshop on Causal Representation Learning.	2023.06 – 2024.05 <i>SNU</i>
Robot Navigation based on Reinforcement Learning <i>Final Year Project</i> - Advisor: Ming Liu (Robotics and Multi-Perception Lab, Robotics Institute) - Title: Map-based Robot Navigation and Path planning with Deep Reinforcement Learning - Proposed ML and RL based algorithm for autonomous navigation in a cluttered environment.	2020.05 – 2021.05 <i>HKUST</i>
Bundleport <i>CTO & Logistics Manager</i> - Created a full-stack web application using Node.js, MySQL, HTML5/CSS3, and JavaScript. - Developed on cloud server using AWS S3, EC2, Elastic Beanstalk and Cloudfront.	2018.05 – 2020.05 <i>HKUST</i>
HKUST Robotics Team <i>Robotics Software Engineer</i> Designed and implemented algorithms for processing data from Camera and LiDAR sensors.	2018.09 – 2018.12 <i>HKUST</i>